

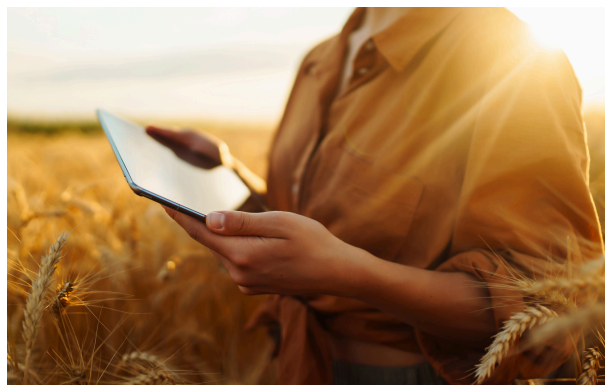
A warm, golden-toned photograph of a person's hands holding a bundle of wheat stalks in a field. The background is softly blurred, showing more wheat and a person's arm. The overall mood is peaceful and agricultural.

A Smart IT Solution for Sustainable Farming and Fair Trade

E Business Strategies
IS - 2112
Group 04

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Problem Domain Analysis

Scenario

In Sri Lanka, when the price of a crops goes up, a few farmers start growing it to make more money. Seeing this, many other farmers also started growing the same crop. When it's ready to harvest, there is too much of that crop in the market. This causes the prices to drop a lot, and farmers can't make enough money to cover their individual costs that it takes to make them. This cycle of high and low prices creates problems for farmers and the market.

Also, the intermediaries make money unethically, reducing profit from the farmers.

SWOT analysis

Strengths

- Sri Lanka has fertile soil and a favorable climate for growing a variety of crops.
- The country produces a wide range of crops, including rice, tea, rubber, and spices.
- Over 30% of the population is employed in agriculture, providing a strong labor force.

Weakness

- Limited use of modern farming technologies in srilankan culture.
- Fragmented land ownership will cause large agricultural projects.
- Highly vulnerable due to climate changes through seasonal like droughts and floods.
- Farmers lack comprehensive market information.

Opportunities

- There is potential to diversify into high-value crops and export-oriented produce.
- Similar to other developed countries, adapting modern agricultural technologies can improve efficiency.
- Improve coordination between farmers and authorities.
- Decentralize current crop buying and distribution system.

Threats

- Rapid global climate changes affect prediction accuracies.
- Sudden Government regulations and price changes.
- Inconsistent production affects food securities and struggles to access credit services for improvements.
- Loss of agricultural knowledge.

Porter Five Forces Analysis

Threat of new entrants

- High initial investments and need of knowledge to startups.
- Government regulations can affect the farming methods.(fertilizer usage, price instability)

Bargaining Power of suppliers

- Rely on mostly government supplied seeds and fertilizers. Government have power over prices.
- Limited number of suppliers.

Bargaining power of buyers

- Limited access to markets due to centralizer distribution system controlled by limited number of parties like Dambulla Economic Centre.
- Consumers are price sensitive, which can limit the quantity of crops sold.

Threat of substitutes

- Imported goods act as substitutes for local products.
- Farmers can switch to different crops if current crop prices fall, leading to the planned seed and fertilizer distribution by the government failing.

Rivalry among competitors

- High competition among farmers, especially during periods of oversupply, can drive prices down.
- The agricultural market is fragmented, with many small scale farmers competing against each other.

PESTEL Analysis

Political

- Insufficient Government subsidies and policies.
- International trade agreements
- Political stability

Economic

- Resources not allocated optimally
- Financial instability
- Limited access to credit and financial services
- Consumer purchasing power depends on national economic growth

Social

- Financial pressure leads to social problems in farmers.
- Price volatility affects low income consumers and it affects their dietary habits.

Technological

- Lack of predicting tools for climate and seasonals
- lack of real time information about market dynamics among farmers

Environmental

- Climate change can significant risk
- Short term focus leads environment degradation
- Inexperienced farmers may overuse fertilizers
- Disposal problems for wastes

Legal

- Insufficient rules governing crop [planning
- Legal issues for the farmers who have not proper ownership for their land of farm
- No proper agreements between farmers and buyers

Strategy Development and Justification

Proposed Strategies

- **Agri Tech Knowledge Hub**
 - Creating an in-app learning center with videos, expert advice, and personalized recommendations helps new farmers get started by making sustainable farming techniques and market trends easier to understand.
- **Government Compliance Alerts**
 - Keeping farmers updated about policy changes, available subsidies, and new regulations through automated notifications helps them avoid legal issues and take advantage of financial opportunities.
- **Micro Investment Platform**
 - Connecting new farmers with investors, NGOs, and loan schemes reduces the financial barriers to entry, letting people start farming with less personal risk.
- **Digital Supplier Marketplace**
 - Building an in-app marketplace that includes private companies and alternative suppliers creates more options for farmers to purchase seeds, fertilizers, and organic products, reducing dependency on government suppliers.
- **Blockchain Supply Chain Tracking**
 - Using blockchain technology to monitor agricultural inputs ensures fair pricing and distribution, preventing manipulation in the supply chain.
- **Bulk Purchasing Groups**
 - Allowing farmers to team up for bulk purchases through the app gives them better negotiating power and reduces costs for everyone involved.
- **Direct to Consumer Sales**
 - Creating a platform where farmers can sell directly to restaurants, hotels, and individual consumers cuts out middlemen and ensures farmers receive fairer prices.
- **AI Price Optimization**
 - Using artificial intelligence to track market prices helps farmers choose the best times and places to sell their crops for maximum profit.
- **Pre Harvest Contracts**
 - Implementing a system for buyers to place orders before harvest gives farmers more certainty about their income and reduces market risk.

- **Local Certification Program**

- Developing a "Grown in Sri Lanka" certification helps local products stand out and appeals to consumers who want to support local agriculture.

- **AI Crop Planning**

- Analyzing market trends with AI helps farmers choose which crops to plant based on predicted demand, reducing uncertainty in decision-making.

- **Cooperative Selling Networks**

- Enabling farmers to form selling cooperatives through the app increases their collective bargaining power when dealing with larger buyers.

- **Harvest Forecasting System**

- Using predictive analytics to anticipate crop yields helps prevent market oversaturation and price crashes during harvest seasons.

- **Value Added Products**

- Encouraging farmers to transform excess produce into processed goods like dried fruits or packaged grains increases profits and reduces waste.

Proposed E-Business Models and Justification

1. Direct to Customer Model

Farmers use a mobile app or website to sell their produce directly to consumers, restaurants, supermarkets, or small retailer bypassing middlemen like the Dambulla Economic Centre.

Justifications

- Solves the Bargaining Power of Buyers Issue,
 - Farmers are often forced to sell at low prices because middlemen control market access. This model removes intermediaries, allowing farmers to set their own fair prices based on demand. As a result, farmers get higher profit margins, and consumers get fresher produce at better prices.
- Addresses Price Sensitivity & Market Access,
 - Consumers are price sensitive, meaning small price differences impact sales. With this model, farmers can offer discounts, subscriptions, loyalty programs to keep customers engaged and online platforms expand the farmer's customer base, reducing reliance on a few centralized buyers.
- Tackles the Threat of New Entrants,
 - High investment & lack of knowledge prevent new farmers from succeeding. A D2C marketplace lowers entry barriers by providing demand data, helping farmers grow what sells best. As a result, Reduces risk by offering pre-orders or subscription-based sales.

Examples

- An app where farmers list their produce, and consumers order directly, with delivery services built in.
- Farm to table platforms like this exist in many regions, reducing dependence on wholesalers.

2. Supply Chain Model

An AI-powered digital supply chain connects farmers to transporters, warehouses, and distributors, optimizing logistics to reduce waste and improve efficiency.

Justifications

- Addresses the Threat of Substitutes,
 - Imported goods often act as substitutes because they are more consistently available and better distributed. A smart supply chain ensures local produce reaches markets on time, reducing reliance on imports.
- Solves Government Control Over Farming Inputs,
 - Farmers depend on government-supplied seeds & fertilizers, which limits flexibility. A supply chain marketplace allows farmers to buy from multiple suppliers, increasing competition and driving costs down.
- Prevents Oversupply & Waste (Competitive Rivalry),
 - Farmers growing the same crops at the same time results in oversupply and price crashes. AI-driven demand forecasting suggests what to plant and when, reducing losses.

Examples

- A real-time logistics platform tracks best transport routes, optimizes cold storage, and suggests best-selling crops based on market data.

3. Revenue Sharing Model

Farmers form digital cooperatives where they sell in bulk to large buyers, such as supermarkets, processors, and exporters, ensuring better prices and fair profit distribution.

Justifications

- Solves Competitive Rivalry Among Farmers,
 - Many small farmers compete in fragmented markets, leading to price wars and reduced profitability. By forming cooperatives, farmers group resources to sell in bulk, securing higher prices from buyers.
- Increases Bargaining Power Against Buyers & Suppliers,
 - Farmers negotiating alone have no power over large buyers. Selling as a group gives them the strength to demand better prices and negotiate bulk deals on seeds & fertilizers.
- Ensures Fair Revenue Distribution,
 - Revenue is split transparently based on each farmer's contribution, ensuring equal benefits. A blockchain-based tracking system can provide automated, fair profit sharing.

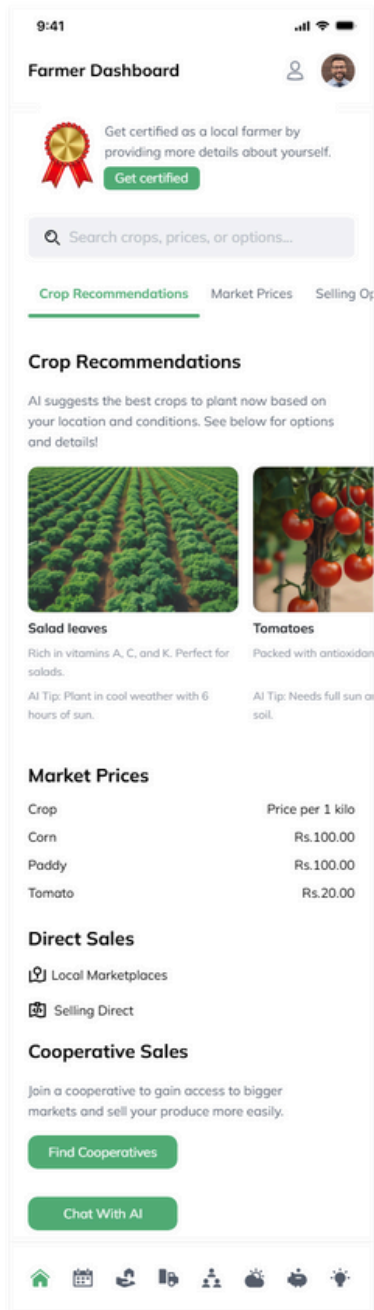
Example

- An app where farmers join cooperative groups, set minimum price limits, and sell to big buyers like supermarkets or exporters.

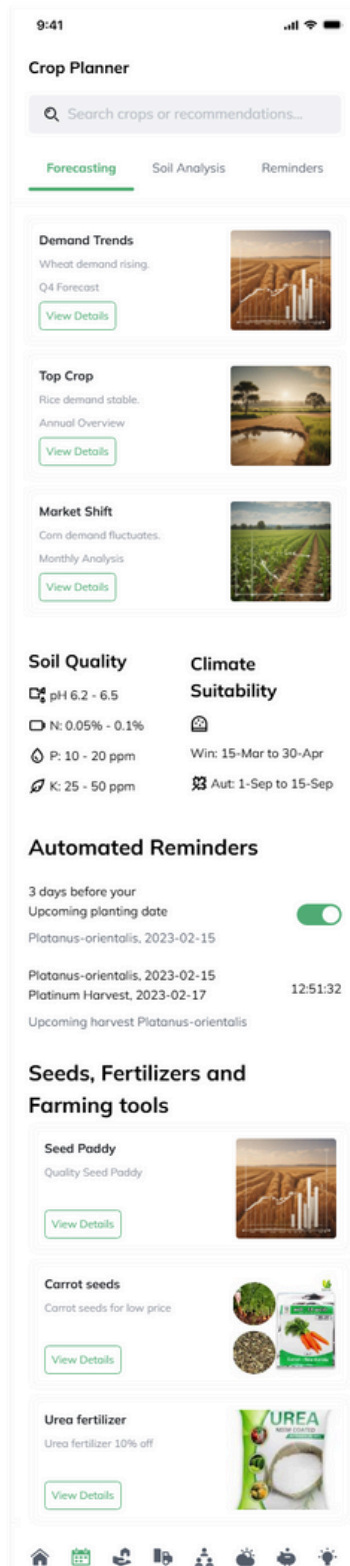
Prototype Screenshots

[Access the prototype system through here.](#)

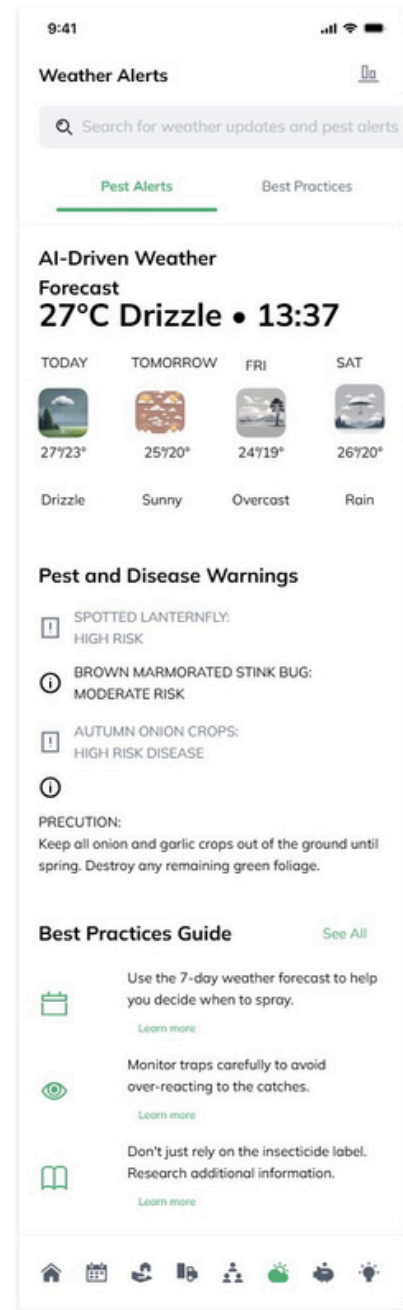
Farmer Dashboard



Crop Planning

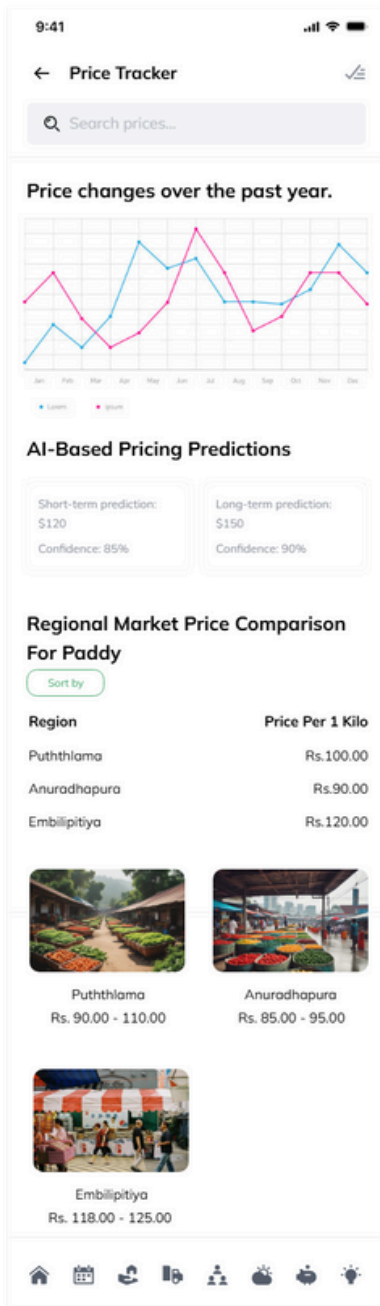


Weather Forecast



Prototype Screenshots

Price Tracking



Knowledge Base

9:41

Cooperative Selling

Search for products or farmers...

Groups Bulk Pricing Revenue Distribution

Green Farmers
Union of paddy farmers in Polgahawela
View Group

Crop Innovators
Collaborate on innovative crop rotation techniques.
View Group

Knowledge Hub
Connect with farmers to share knowledge and resources.
View Group

Market Collective
Promote local produce through cooperative markets.
View Group

Create a New Group

Search for groups

Farmer's Group
Join our farmer's market group to share and discover local produce and farming tips.
Join Now

Farmer's Group
Join our farmer's market group to share and discover local produce and farming tips.
Join Now

Farmer's
Join our farmer's market group to share and discover local produce and farming tips.
Join Now

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Cooperative Selling

9:41

Knowledge Base

Search articles...

Articles Government Compliance & Policy Updates

Crops Livestock Equipment Sustainable Practices

Understanding the Future of Agriculture: A Closer Look at Precision Farming

Explore how precision farming techniques are revolutionizing agriculture and helping farmers increase efficiency and yields. Full article on AgriTech News.

12 May 2023

5 Innovative Tools Every Modern Farmer Should Consider

Learn about the top five innovative tools that can help farmers enhance productivity and streamline operations in their daily tasks.

10 May 2023

Crop Rotation: The Key to Sustainable Farming Practices

Discover the benefits of crop rotation and how it can lead to healthier soils and improved crop yields over time.

05 May 2023

The Rise of Vertical Farming: How Urban Agriculture is Changing Our Cities

Explore the concept of vertical farming and its potential to transform urban landscapes while providing fresh produce to city dwellers.

28 Apr 2023

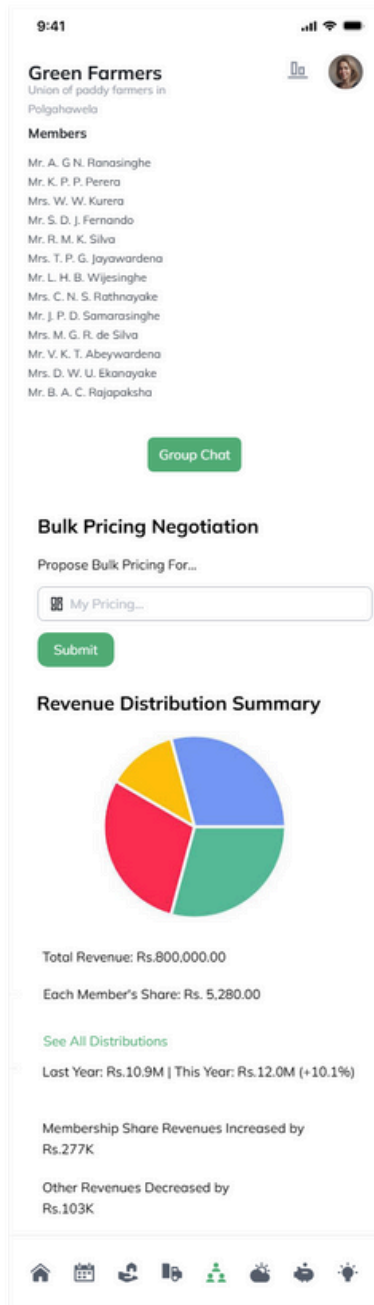
Exploring Digital Tools in Agriculture: A Farmer's Perspective

By Aidan R. Sweeney | 10/1/2023

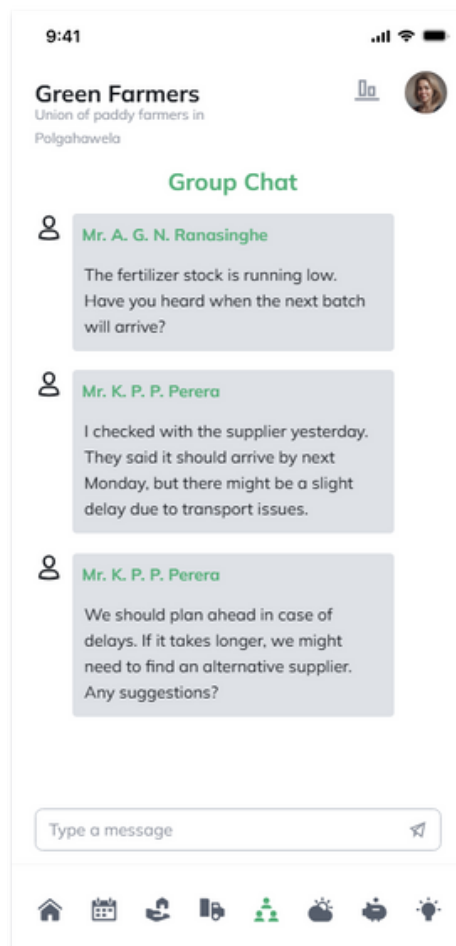
Home Calendar Search Groups Market Knowledge Hub News Settings

Prototype Screenshots

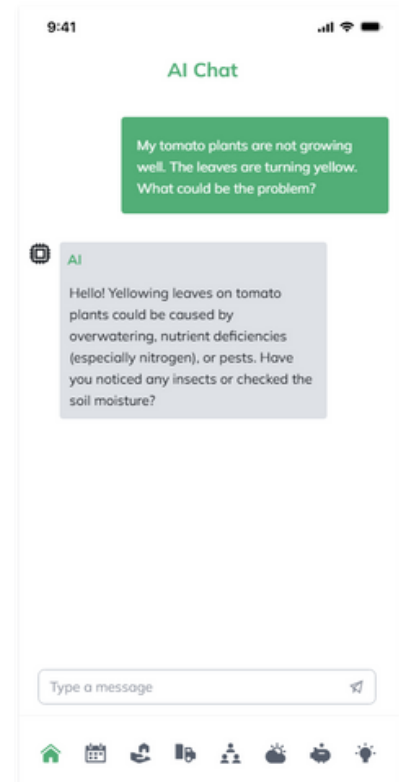
Farmers Group



Group Chat

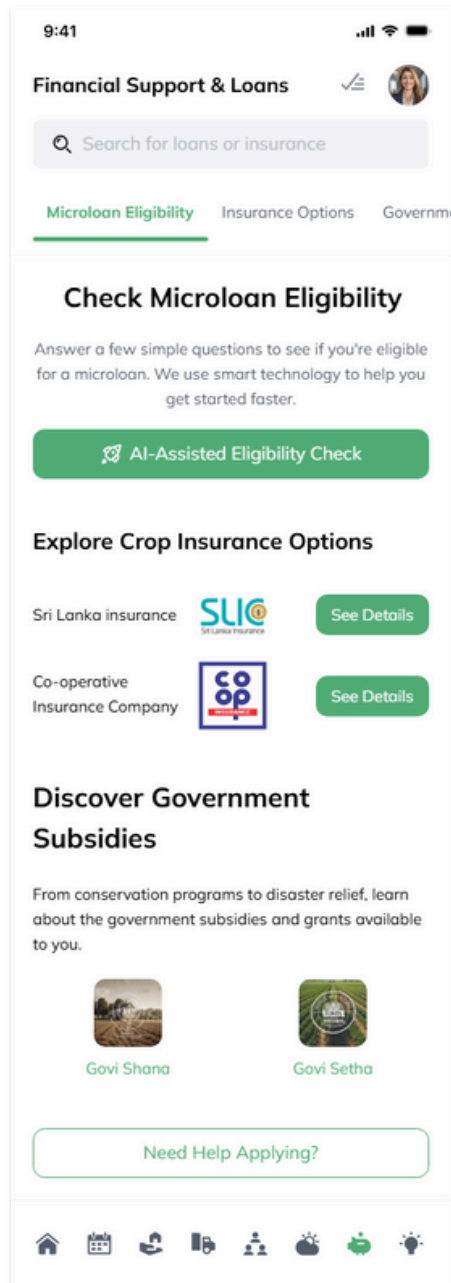


AI Chat

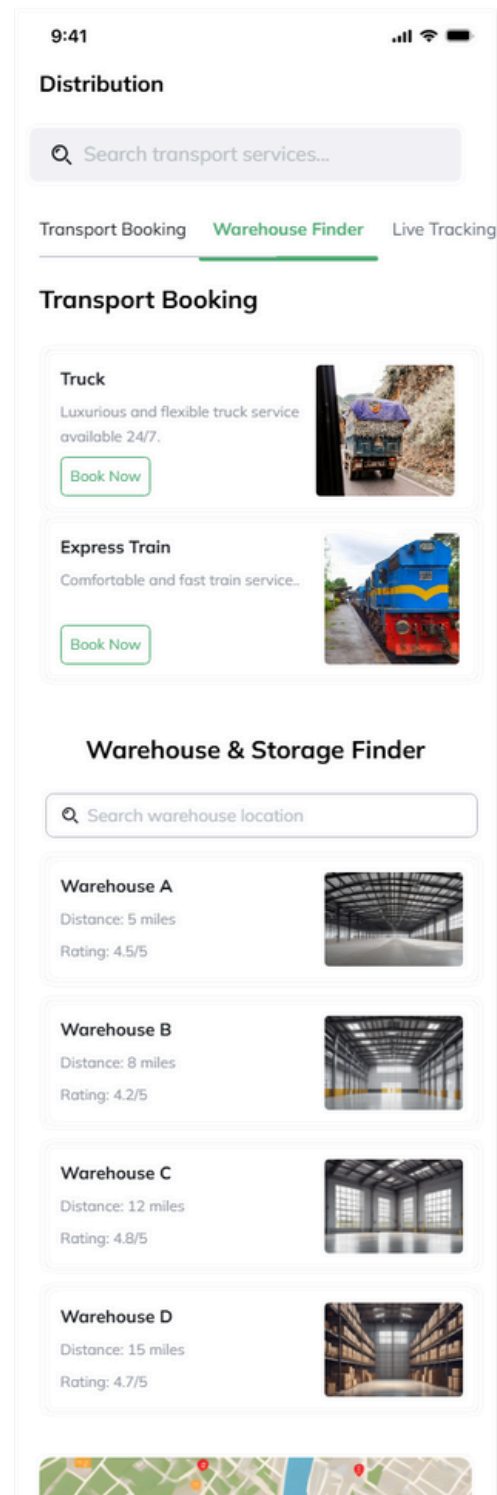


Prototype Screenshots

Financial Support & Loans



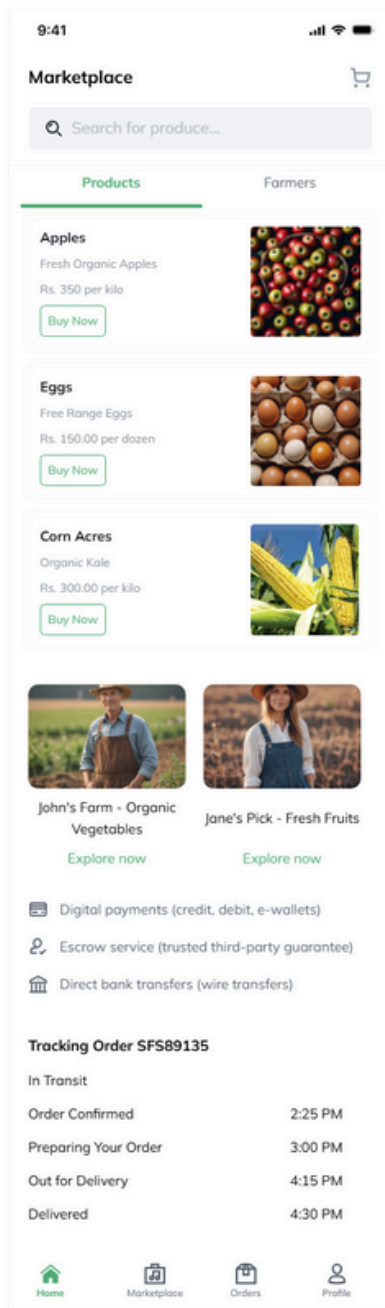
Distribution



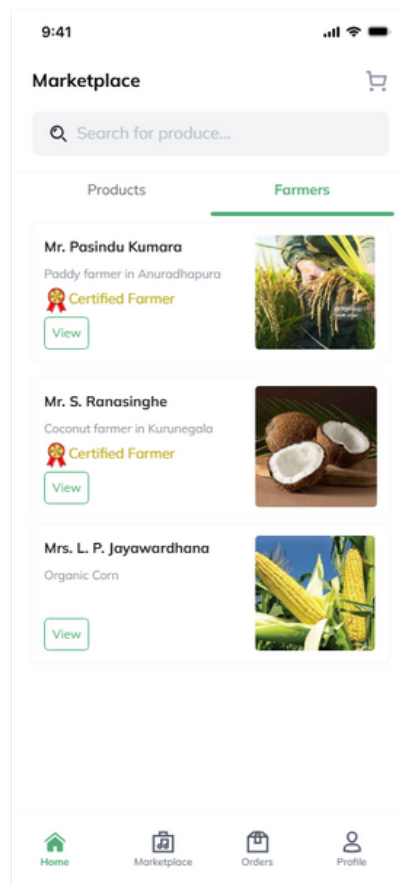
Prototype Screenshots

Buyer's Views

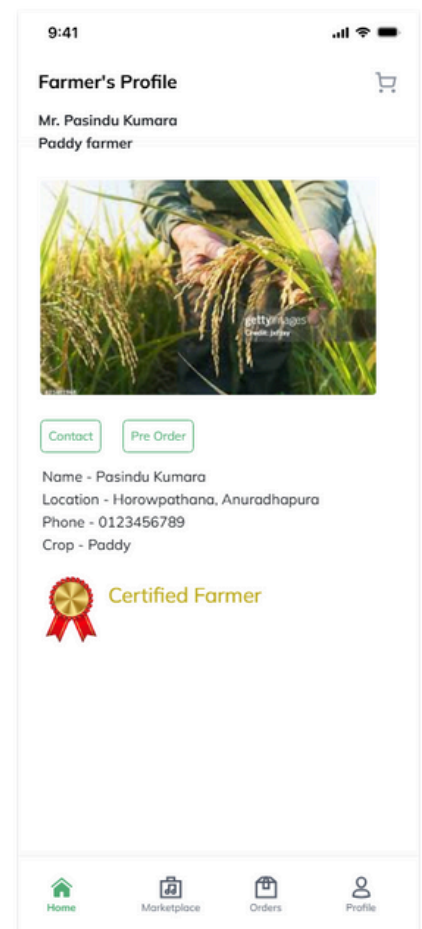
Products



Farmers



Farmers Profile



For viewing all the UI s - <https://app.visily.ai/projects/0ddc4446-ba21-4206-8c61-c92118df85b9/boards/1691227>

E -Transformation Methodology

To tackle the challenges of oversupply, price drops, and exploitation by intermediaries in the agriculture sector, the eTransformation methodology provides a thorough strategy for reconfiguring business processes using digital technology. Here are some usages.

Market Linkage through digital platform

- eTransformation can be used to develop a digital platform (a mobile app) where farmers can directly connect with consumers, retailers, or cooperatives. This results in the elimination of reliance on middlemen and ensures farmers get better prices for their produce. Through the app, farmers can list their products, set prices, and engage in direct transactions with buyers.

Supply and demand forecasting

- Using predictive analytics and AI based systems to forecast market demand, enabling farmers to grow crops based on anticipated market needs which reduces overproduction. Through these systems, farmers get accurate predictions for demand trends, helping them make informed decisions on crop planting.

Crop sharing network

- Enabling the option for farmers to collaborate and share resources, including equipment, labor, and knowledge, to reduce individual costs and increase efficiency. Farmers are able to join cooperative networks, share resources, and mitigate the risk of price fluctuations by growing a diverse range of crops collectively.

Automated logistics and distribution

- Enabling the efficiency of distributing goods directly from farmers to retailers, reducing delays and unnecessary costs imposed by middlemen. Through the app, farmers can manage the inventory, distribution, and delivery processes, ensuring timely and cost effective deliveries.

Data Driven Pricing mechanisms

- Farmers can use data driven pricing to set competitive rates for their products, optimizing their revenue without the influence of middlemen.

Conclusion

The agricultural industry faces significant challenges, including market inefficiencies, price instability, and the power of middlemen. Farmers often struggle with low bargaining power, high competition, and limited access to direct markets, which negatively impact their profitability.

Our proposed IT solution, “AgriSmart” leverages technology to address these challenges and create a sustainable, farmer centric ecosystem.

By implementing a Direct to Customer model, farmers can sell produce directly to consumers, bypassing intermediaries and securing fair prices. The Supply Chain Model optimizes logistics and distribution, ensuring efficiency and reducing waste, while the Revenue Sharing Model empowers farmers to sell in bulk, negotiate better deals, and equitably share profits. Additionally, AI powered crop planning, real time price tracking, and digital cooperatives enhance decision making and market adaptability.

Together, these models form a comprehensive ebusiness solution that not only increases farmer profitability but also strengthens the agricultural supply chain. By integrating technology, data analytics, and digital connectivity, “AgriSmart” creates a fair, transparent, and sustainable agricultural marketplace. This initiative has the potential to transform the farming industry, ensuring that farmers receive equitable compensation for their hard work while promoting economic stability and food security.

“AgriSmart” is more than just a technological solution, it is a revolution in modern farming, creating the way for a more resilient, efficient and farmer friendly agricultural sector.

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